

Molindone

Therapeutics

Brands

- Moban
- *see index for additional brand names*

Generic?

Yes

Class

- Conventional antipsychotic (neuroleptic, dopamine 2 antagonist)

Commonly Prescribed for

(bold for FDA approved)

- **Schizophrenia (no longer available in the USA)**
- Other psychotic disorders
- Bipolar disorder

How the Drug Works

- Blocks dopamine 2 receptors, reducing positive symptoms of psychosis

How Long Until It Works

- Psychotic symptoms can improve within 1 week, but it may take several weeks for full effect on behavior

If It Works

- Most often reduces positive symptoms in schizophrenia but does not eliminate them
- Most schizophrenic patients do not have a total remission of symptoms but rather a reduction of symptoms by about a third
- Continue treatment in schizophrenia until reaching a plateau of improvement
- After reaching a satisfactory plateau, continue treatment for at least a year after first episode of psychosis in schizophrenia
- For second and subsequent episodes of psychosis in schizophrenia, treatment may need to be indefinite
- Reduces symptoms of acute psychotic mania but not proven as a mood stabilizer or as an effective maintenance treatment in bipolar disorder
- After reducing acute psychotic symptoms in mania, switch to a mood stabilizer and/or an atypical antipsychotic for mood stabilization and maintenance

If It Doesn't Work

- Consider trying one of the first-line atypical antipsychotics

- Consider trying another conventional antipsychotic
- If 2 or more antipsychotic monotherapies do not work, consider clozapine

Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation of conventional antipsychotics has not been systematically studied
- Addition of a mood-stabilizing anticonvulsant such as valproate, carbamazepine, or lamotrigine may be helpful in both schizophrenia and bipolar mania
- Augmentation with lithium in bipolar mania may be helpful
- Addition of a benzodiazepine, especially short-term for agitation

Tests

* Since conventional antipsychotics are frequently associated with weight gain, before starting treatment, weigh all patients and determine if the patient is already overweight (BMI 25.0–29.9) or obese (BMI >30)

- Before giving a drug that can cause weight gain to an overweight or obese patient, consider determining whether the patient already has pre-diabetes (fasting plasma glucose 100–125 mg/dL), diabetes (fasting plasma glucose >126 mg/dL), or dyslipidemia (increased total cholesterol, LDL cholesterol, and triglycerides; decreased HDL

cholesterol), and treat or refer such patients for treatment, including nutrition and weight management, physical activity counseling, smoking cessation, and medical management

- * Monitor weight and BMI during treatment
- * Consider monitoring fasting triglycerides monthly for several months in patients at high risk for metabolic complications and when initiating or switching antipsychotics
- * While giving a drug to a patient who has gained >5% of initial weight, consider evaluating for the presence of pre-diabetes, diabetes, or dyslipidemia, or consider switching to a different antipsychotic
- Should check blood pressure in the elderly before starting and for the first few weeks of treatment
- Monitoring elevated prolactin levels of dubious clinical benefit
- Patients with low white blood cell count (WBC) or history of drug-induced leukopenia/neutropenia should have complete blood count (CBC) monitored frequently during the first few months and molindone should be discontinued at the first sign of decline of WBC in the absence of other causative factors
- Patients should be monitored periodically for the development of abnormal movements of tardive dyskinesia, using neurological exam and the AIMS (Abnormal Involuntary Movement Scale)

Side Effects

How Drug Causes Side Effects

- Acute blockade of dopamine 2 receptors in the striatum can cause drug-induced parkinsonism, dystonia, or akathisia
- Chronic blockade of dopamine 2 receptors in the striatum can cause tardive dyskinesia
- By blocking dopamine 2 receptors in the pituitary, it can cause elevations in prolactin
- By blocking dopamine 2 receptors excessively in the mesocortical and mesolimbic dopamine pathways, especially at high doses, it can cause worsening of negative and cognitive symptoms (neuroleptic-induced deficit syndrome)
- Blocking muscarinic cholinergic receptors can cause dry mouth, blurred vision, urinary retention, constipation, and paralytic ileus
- Antihistaminic actions may cause sedation, weight gain
- Blocking alpha 1 adrenergic receptors can cause dizziness, hypotension, and syncope
- Mechanism of weight gain and increased incidence of diabetes and dyslipidemia with atypical antipsychotics is unknown, and risks vary based on the particular agent

Notable Side Effects

- * Neuroleptic-induced deficit syndrome
- * Akathisia

- * Drug-induced parkinsonism
- * Tardive dyskinesia
- * Risk of potentially irreversible involuntary dyskinetic movements may increase with cumulative dose and treatment duration
- * Galactorrhea, amenorrhea
- Sedation
- Dry mouth, constipation, vision disturbance, urinary retention
- Hypotension, tachycardia

Life-Threatening or Dangerous Side Effects

- Rare neuroleptic malignant syndrome (NMS) may cause hyperpyrexia, muscle rigidity, delirium, and autonomic instability with elevated creatine phosphokinase, myoglobinuria (rhabdomyolysis), and acuterenal failure
- Rare leukopenia
- Rare seizures
- As a class, antipsychotics are associated with an increased risk of death and cerebrovascular events in elderly patients with dementia; not approved for treatment of dementia-related psychosis

Weight Gain



- Reported but not expected

Sedation



- Many experience and/or can be significant in amount
- Sedation is usually transient

What to Do About Side Effects

- Wait
- Wait
- Wait
- Anticholinergics may reduce drug-induced parkinsonism when present
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) may reduce akathisia
- Reduce the dose
- For sedation, give at night
- Switch to an atypical antipsychotic
- Weight loss, exercise programs, and medical management for high BMIs, diabetes, dyslipidemia
- Metformin may help prevent or reverse antipsychotic-induced weight gain

Best Augmenting Agents for Side Effects

- Benztropine, trihexyphenidyl, or amantadine for drug-induced parkinsonism
- Beta blockers, benzodiazepines, or 5HT_{2A} antagonists (e.g., mirtazapine, cyproheptadine) for akathisia
- Valbenazine or deutetrabenazine for tardive dyskinesia
- Many side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 40–100 mg/day in divided doses

Dosage Forms

- Tablet 5 mg, 10 mg, 25 mg scored, 50 mg scored, 100 mg scored
- Liquid 20 mg/mL

How to Dose

- Initial 50–75 mg/day; increase to 100 mg/day after 3–4 days; maximum 225 mg/day



Dosing Tips

- Very short half-life, but some patients may only require once daily dosing

- Other patients may do better with 3 or 4 divided doses daily
- Patients receiving atypical antipsychotics may occasionally require a “top-up” of a conventional antipsychotic to control aggression or violent behavior
- Rather than raise the dose above normal dosing in acutely agitated patients requiring acute antipsychotic actions, consider short-term (one dose or more) augmentation with a benzodiazepine or another antipsychotic, especially intramuscularly
- Rather than raise the dose above normal dosing in partial responders, consider augmentation with a mood-stabilizing anticonvulsant, such as valproate, topiramate, or lamotrigine
- Children and elderly should generally be dosed at the lower end of the dosage spectrum
- Treatment should be suspended if absolute neutrophil count falls below $1000/\text{mm}^3$

Overdose

- Deaths have occurred; drug-induced parkinsonism, sedation, hypotension, respiratory depression, coma

Long-Term Use

- Should periodically reevaluate long-term usefulness in individual patients, but treatment may need to continue for many years

Habit Forming

- No

How to Stop

- Slow down-titration (over 6–8 weeks), especially when simultaneously beginning a new antipsychotic while switching (i.e., cross-titration)
- Rapid discontinuation may lead to rebound psychosis and worsening of symptoms
- If antiparkinson agents are being used, they should be continued for a few weeks after molindone is discontinued

Pharmacokinetics

- Half-life approximately 1.5 hours



Drug Interactions

- Additive effects may occur if used with CNS depressants
- Some patients taking a neuroleptic and lithium have developed an encephalopathic syndrome similar to neuroleptic malignant syndrome
- Molindone tablets contain calcium sulfate, which may interfere with absorption of phenytoin sodium or tetracyclines
- Combined use with epinephrine may lower blood pressure

- May increase the effects of antihypertensive drugs

Other Warnings/Precautions

- If signs of neuroleptic malignant syndrome develop, treatment should be immediately discontinued
- Liquid molindone contains sodium metabisulfite, which may cause allergic reactions in some people, especially in asthmatic people
- Use cautiously in patients with alcohol withdrawal or convulsive disorders because of possible lowering of seizure threshold
- Antiemetic effect can mask signs of other disorders or overdose
- Do not use epinephrine in event of overdose as interaction with some pressor agents may lower blood pressure
- Use cautiously in patients with glaucoma, urinary retention
- Observe for signs of ocular toxicity (pigmentary retinopathy, lenticular pigmentation)
- Use only with caution if at all in Parkinson's disease or Lewy body dementia
- Antipsychotics may cause somnolence, postural hypotension, motor and sensory instability, which may lead to falls; for patients with diseases, conditions, or medications that could exacerbate these effects, complete fall risk assessments when initiating antipsychotic treatment, and recurrently for patients on long-term antipsychotic therapy

- As with any antipsychotic, use with caution in patients with history of seizures

Do Not Use

- If patient is in a comatose state or has CNS depression
- If there is a proven allergy to molindone

Special Populations

Renal Impairment

- Should receive initial lower dose

Hepatic Impairment

- Should receive initial lower dose

Cardiac Impairment

- Use with caution

Elderly

- Should receive initial lower dose
- Although conventional antipsychotics are commonly used for behavioral disturbances in dementia, no agent has been approved for treatment of elderly patients with behavioral symptoms of dementia such as agitation

- Elderly patients with dementia-related psychosis treated with antipsychotics are at an increased risk of death compared to placebo, and also have an increased risk of cerebrovascular events

Children and Adolescents

- Safety and efficacy not well established
- Generally consider second-line after atypical antipsychotics

Pregnancy

- Controlled studies have not been conducted in pregnant women
- Animal studies have not shown adverse effects
- There is a risk of abnormal muscle movements and withdrawal symptoms in newborns whose mothers took an antipsychotic during the third trimester; symptoms may include agitation, abnormally increased or decreased muscle tone, tremor, sleepiness, severe difficulty breathing, and difficulty feeding
- Psychotic symptoms may worsen during pregnancy and some form of treatment may be necessary
- Atypical antipsychotics may be preferable to conventional antipsychotics or anticonvulsant mood stabilizers if treatment is required during pregnancy

Breast Feeding

- Unknown if molindone is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Some patients benefit from molindone's sedative properties

Potential Disadvantages

- Patients with tardive dyskinesia

Primary Target Symptoms

- Positive symptoms of psychosis
- Motor and autonomic hyperactivity
- Violent or aggressive behavior

Pearls

- After having been produced and subsequently discontinued by Core Pharma in 2015–2017, molindone is available again from Epic Pharma effective December, 2018
- * May cause less weight gain than some other antipsychotics

- Not shown to be effective for behavioral problems in mental retardation
- Patients have very similar antipsychotic responses to any conventional antipsychotic, which is different from atypical antipsychotics where antipsychotic responses of individual patients can occasionally vary greatly from one atypical antipsychotic to another
- Patients with inadequate responses to atypical antipsychotics may benefit from a trial of augmentation with a conventional antipsychotic such as molindone or from switching to a conventional antipsychotic such as molindone
- However, long-term polypharmacy with a combination of a conventional antipsychotic such as molindone with an atypical antipsychotic may combine their side effects without clearly augmenting the efficacy of either
- Although a frequent practice by some prescribers, adding two conventional antipsychotics together has little rationale and may reduce tolerability without clearly enhancing efficacy

Suggested Reading

Bagnall A , Fenton M , Kleijnen J , Lewis R. Molindone for schizophrenia and severe mental illness . *Cochrane Database Syst Rev* 2007 ;**24** (1):CD002083 .

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Huhn M , Nikolakopoulou A , Schneider-Thoma J , et al. Comparative efficacy and tolerability of 32 oral antipsychotics for the acute treatment of adults with multi-episode schizophrenia: a systematic review and network meta-analysis . *Lancet* 2019 ;**394** :939–51 .

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Owen RR Jr , Cole JO . Molindone hydrochloride: a review of laboratory and clinical findings . *J Clin Psychopharmacol* 1989 ;**9** (4):268–76 .

Nalmefene

Therapeutics

Brands

- Selincro
- *see index for additional brand names*

Generic?

Yes (not for oral)

Class

- Neuroscience-based Nomenclature: opioid receptor antagonist (O-RAn)
- Alcohol dependence treatment; mu and delta opioid receptor antagonist and kappa opioid receptor partial agonist

Commonly Prescribed for

(bold for FDA approved)

- Reduction of alcohol consumption in patients with alcohol dependence who have a high drinking risk level

How the Drug Works

- Reduces alcohol consumption through modulation of opioid systems, thereby reducing the reinforcing effects of alcohol
- Blockade of mu opioid receptors prevents the pleasurable effects of alcohol, whereas modulation of the kappa opioid receptors may reduce dysphoria associated with alcohol withdrawal

How Long Until It Works

- Can begin working immediately and can be used as needed

If It Works

- Reduces alcohol consumption by diminishing reinforcing properties of alcohol (rewarding effects, cravings)

If It Doesn't Work

- Evaluate for and address contributing factors
- Consider switching to another agent



Best Augmenting Combos for Partial Response or Treatment Resistance

- Augmentation with behavioral, educational, and/or supportive therapy in groups or as an individual is key to successful treatment

Tests

- None for healthy individuals, although baseline liver function testing, usually obtained anyway for managing alcohol dependence, may be useful

Side Effects

How Drug Causes Side Effects

- Blockade of mu opioid receptors

Notable Side Effects

- Nausea, vomiting
- Dizziness, insomnia, headache



Life-Threatening or Dangerous Side Effects

- Confusion, rare hallucinations

Weight Gain



- Reported but not expected
- May cause weight loss

Sedation



- Occurs in significant minority

What to Do About Side Effects

- Wait
- Switch to another agent

Best Augmenting Agents for Side Effects

- Switching to another agent may be more effective since most side effects cannot be improved with an augmenting agent

Dosing and Use

Usual Dosage Range

- 18 mg/day as needed

Dosage Forms

- Tablet 18 mg

How to Dose

- After an initial visit, patient should keep a record of alcohol consumption for 2 weeks; patients who continue to have a high drinking risk level during those 2 weeks can be initiated on nalmefene
- Nalmefene is taken as needed: on each day the patient perceives a risk of drinking alcohol, 18 mg should be taken 1–2 hours before the anticipated time of drinking

- If the patient has already begun drinking, 18 mg should be taken as soon as possible
- Maximum dose is 18 mg/day
- Patient should be opioid-free for 7–10 days prior to initiating treatment



Dosing Tips

- Can be taken with or without food
- Providing educational materials and counseling in combination with nalmefene treatment can increase the chances of success
- Nalmefene tablet should not be chewed or crushed, as nalmefene may cause skin sensitization when in direct contact with skin

Overdose

- Limited experience

Long-Term Use

- Has been evaluated in trials for up to 1 year

Habit Forming

- No

How to Stop

- Taper not necessary

Pharmacokinetics

- Extensively metabolized by the liver
- Terminal half-life is 12.5 hours



Drug Interactions

- Increased depressive effects, particularly respiratory depression, have occurred when taken with other CNS depressants; consider dose reduction of either or both when taken concomitantly
- UGT2B7 inhibitors (e.g., diclofenac, fluconazole, medroxyprogesterone acetate, meclofenamic acid) may increase nalmefene levels
- UGT2B7 inducers (e.g., dexamethasone, phenobarbital, rifampicin, omeprazole) may decrease nalmefene levels



Other Warnings/Precautions

- To prevent withdrawal in patients dependent on opioids, patients must be opioid-free for at least 7–10 days prior to initiating treatment
- Individuals receiving nalmefene who require pain management with opioid analgesia may need a higher dose than usual and may experience deeper and more prolonged respiratory depression; pain

management with non-opioid or rapid-acting opioid analgesics is recommended if possible

- Nalmefene should be temporarily discontinued 1 week prior to anticipated use of opioids (e.g., opioid analgesia during elective surgery)
- Use caution when using products containing opioids (e.g., cough medicines)
- Risk of respiratory depression is increased with concomitant use of CNS depressants

Do Not Use

- If patient is taking opioid analgesics
- If patient is currently dependent on opioids or is in acute opiate withdrawal
- If patient has severe renal or hepatic impairment
- If patient has a recent history of acute alcohol withdrawal syndrome
- If patient has galactose intolerance, Lapp lactase deficiency, or glucose-galactose malabsorption
- If there is a proven allergy to nalmefene

Special Populations

Renal Impairment

- Dose adjustment not necessary for mild to moderate impairment
- Not recommended for use in severe impairment

Hepatic Impairment

- Dose adjustment not necessary for mild to moderate impairment
- Not recommended for use in severe impairment

Cardiac Impairment

- Not studied

Elderly

- Limited data available



Children and Adolescents

- Safety and efficacy have not been established



Pregnancy

- Controlled studies have not been conducted in pregnant women
- Some animal studies have shown adverse effects
- Pregnant women needing to stop drinking may consider behavioral therapy before pharmacotherapy
- Not generally recommended for use during pregnancy, especially during first trimester

Breast Feeding

- Unknown if nalmefene is secreted in human breast milk, but all psychotropics are assumed to be secreted in breast milk
- * Recommended either to discontinue drug or bottle feed

The Art of Psychopharmacology

Potential Advantages

- Individuals who are not ready to abstain completely from alcohol

Potential Disadvantages

- Individuals whose goal is immediate abstinence

Primary Target Symptoms

- Alcohol dependence

Pearls

- Nalmefene was originally used as a parenteral agent to reverse the opioid agonist effects of opioid anesthesia or in opioid overdose
- In Europe, nalmefene is approved for reduction of alcohol consumption without previous detoxification
- Nalmefene is intended for patients with the goal of reduced-risk drinking (i.e., 3–4 drinks per day in men, maximum 16 drinks per week; 2–3 drinks per day in women, maximum 12 drinks per week);